

TOTAL COST OF OWNERSHIP				
Number of Vehicles	1			
Life of Vehicle (years)	10			
Annual drive distance (KM)	23725			
Total Distance Travel (KM)	237250			
LIB Cost (USD/kWh)	156			
USD to INR	71			
VRLA Cost (USD/kWh)	100			
General Inputs	e-2W (Lead acid battery)	e-2W (Li-ion small battery)	e-2W (Li-ion large battery)	Petrol-2W
Discount rate (%)	10.00%	10.00%	10.00%	10.00%
Resale rate (%)	10.00%	11.00%	14.00%	6.00%
Capital Cost	e-2W (Lead acid battery)	e-2W (Li-ion small battery)	e-2W (Li-ion large battery)	Petrol-2W
Total Vehicle Cost (₹)	88,485.17	108,145.81	192,205.38	98,049.29
Resale Value (₹)	8,848.52	11,896.04	26,908.75	5,882.96
Total Charging Infrastructure Cost (₹)	-	-	-	-
Misc. Cost (₹)	-	-	-	-
Financial Incentive (₹)	-	17,000.00	29,000.00	-
Total Capital Cost (₹) excl. Resale Value	88,485.17	91,145.81	163,205.38	98,049.29
Annual Operational Cost	e-2W (Lead acid battery)	e-2W (Li-ion small battery)	e-2W (Li-ion large battery)	Petrol-2W
Staff Cost (₹)	-	-	-	-
Maintenance Cost (₹)	2,480.00	2,589.00	3,620.00	3,920.00
Battery Replacement Cost (₹)*	23,004.00	3,876.60	2,658.24	-
Average Fuel Cost (₹)	3,650.00	3,329.82	3,216.95	49,947.37
Misc. Cost (₹)	-	-	-	-
Total Operational Cost (₹)	29,134.00	9,795.42	9,495.19	53,867.37
Total Operational Cost for life of 2W	291,340.00	97,954.25	94,951.89	538,673.68
Average TCO	1.34	0.85	1.29	2.05

be higher for some more time owing to reliance on imports and battery costs. The industry should endeavor to investigate other ways of subsidising their products to reach a larger customer base. The government can also further bolster FAME by offering performance-based incentives.

Improved telemetry: Currently, most EVs employ some form of telemetry that is presented in an assimilable form to the rider. However, there is still lot of scope for improving the applications of this data such as for taking intelligent braking and acceleration decisions on the road, customising the suspension for known bad roads, etc. This telemetry can also be used to tailor-make software and services that cater to Indian road conditions and weather patterns.

Proper charging infrastructure: Concerns of lack of charging infrastructure probably ranks only second to

range anxiety in hindering wide adoption of EVs in India. India currently has only about 1,800 public charging stations. The government has been encouraging power companies and even oil marketing companies (OMCs) to setup adequate charging infrastructure. OMCs have collectively pledged to setup 22,000 charging stations across the country. However, a lot more needs to be done and private players also need to step-in to effectively cover a vast country like India.

Increased R&D investments: Tackling a nascent but potential market requires considerable investments in R&D. The diverse landscape and weather patterns in India provide ample opportunities for OEMs to research and publish whitepapers on various aspects such as battery quality, effect of temperature and humidity on battery life, explore recycling and reusing of ageing batteries, tailoring

EV components to be resistant to typical usage conditions, etc.

Scrappage policy: Vehicle recycling is currently an unorganised and unregulated sector in India. Millions of obsolete vehicles litter Indian cities with no way to properly salvage parts. Workers in these scrapyards often work in hazardous conditions. This also contributes to a lot of environmental pollution. States such as Delhi have a scrappage policy that incentivises those who scrap their ICE for an EV. Alongside government support, OEMs will also have to price their products in a way that is lucrative enough to motivate prospective customers to scrap their ICEs. This also opens a whole new business opportunity given the sheer number of vehicles on Indian roads. A nationwide scrapping policy like that of the EU or Japan is a dire need of the hour in India.

TOTAL COST OF OWNING AN EV

The main driving force for EV adoption is the potential to save huge on fuel costs. However, the joy of saving on fuel costs often dampens upon looking at the upfront costs of purchasing an EV. A recent report from RMI-India calculates significant savings in total cost of ownership (TCO) for EV four-wheelers compared to diesel, petrol, and CNG equivalents but only if adequately incentivised.

As you can see in the table above, even though the upfront cost for a two-wheeler EV is higher, the average TCO comes to for an EV with a small Li-ion battery is just 0.85 over a 10-year vehicle lifetime compared to 2.05 for the petrol equivalent even after factoring battery replacement costs. Even a two-wheeler EV with a large Li-ion battery or lead-acid battery offers significant savings over its ICE equivalent. It is further evident that financial incentives play a huge role in lowering the EV entry barrier and reducing overall TCO.

A subscription model could be a possible solution that allows the customer to use the EV for a designated number of hours or kms travelled can help lower total initial costs and alleviates worries of financing options. 